MILITARY SPECIFICATION SHEET

ELECTRON TUBE, NEGATIVE GRID (MICROWAVE)
TYPE 7289

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, planar, ceramic and metal
See figure 1
Mounting position: Any
Weight: 2.5 ounces nominal

ABSOLUTE RATINGS:

Parameter: / Unit:	F1 GHz	Ef V (see note 2)	Eb Vdc	eb v	epy kv	Ec Vdc	Ik mAdc	Ib mAdc	ib a	Ic mAdc	ic a
MAXIMUM:											
CW osc	2.5	6.0±5%	1,000	---	---	-150	125	---	---	50	---
Anode mod CW osc or amp	2.5	6.0±5%	---	1,200 (see note 3)	---	-150	100	---	---	50	---
Anode pulsed osc or amp (Note 1)	3.0	6.0±5%	---	---	3.5	---	---	---	3	---	1.8
TEST CONDITIONS:	---	6.0	1,000	---	---	Adj	---	100	---	---	---

Parameter: / Unit:	tp μs	Du ---	Pp W (see note 4)	Pg W (see note 6)	tk Sec (min)	TE °C (see note 7)	T (anode shank) °C (see note e of figure 1)	Cooling ---	Barometric pressure, reduced mmHg
MAXIMUM:									
CW osc	---	---	100	2	60	300	300	---	54
Anode mod CW osc or amp	---	---	100 (see note 5)	2	60	300	300	---	54
Anode pulsed osc or amp (see note 1)	3	0.0025	27	---	60	300	300	---	---
TEST CONDITIONS:	---	---	---	---	300	---	---	Note 8	---

GENERAL:

Qualification - Required

(E) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1	15							
(E) 1211	Insulation of electrodes	-	Eb = Ek = 0; Ec = -500 Vdc	0.65	II	R	50	---	Meg
1236	Power oscillation	13, 16	F = 2.5 GHz (min); Ebb = 1,000 Vdc; Ib = 90 mAdc; Ef = 5.0 V	0.65	II	Po	15	---	W
1261	Electrode voltage (1) (grid)	17		0.65	II	Ec	-2.0	-7.0	Vdc
1266	Total grid current	17		0.65	II	Ic	---	-8	μAdc
(E) 2212	Pulse emission (1)	-	Eb = 145 Vdc; Ec = -10 to -150 Vdc; egk/ik = 750 ma; Ef = 5.4 V; prr = 11 ± 1 pps; tp = 4,500 μs (min); tr = tf = 25 μs; slope = 0.5 percent; ripple = 0.1 percent	0.65	II	Δik	---	120	ma
1372	Current division (long pulse)	-	Pulse emission (1), except egk/ib = 460 ma; Ef = 6.0 V	0.65	II	egk ic	14 200	22 320	v ma
1301	Heater current	-		0.65	II	If	0.90	1.05	A
	Quality conformance inspection, part 2								
1236	Power oscillation (pulse)	13, 19	F = 3.0 GHz (min); epy = 3.5 kv; Ec = -1.5 V (min); Rg/Ib = 7.5 mA (max); Ic = 4.5 mA (max); Ef = 5.8 V	---	---	Po	4	---	W
1331	Direct-interelectrode capacitance	13, 20	No voltages	---	---	Cin Cgp Cout	5.6 1.95 ---	7.0 2.15 0.035	pF pF pF
---	Resonance	13, 21	No voltages	---	---	---	---	---	---
---	Grid distortion	13, 22		---	---	ΔF	+1	-7	MHz
1261	Electrode voltage (2) (grid)	-	Ec/Ib = 1.0 mAdc	---	---	Ec	---	-25	Vdc
---	Grid arc	-	See figure 2	---	---	---	---	---	---
(E) 2212	Pulse emission (2)	-	Pulse emission (1) except Ef = 6.0 V	---	---	Δik	---	60	ma

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3								
---	Life test (1)	11	Group S (monthly); F = 2.0 GHz (min); Ebb = 1,000 Vdc; Ib = 90 mAdc; Ef = 5.0 V; t = 200 hours	---	---	---	---	---	---
---	Life-test (1) end points:								
1236	Power oscillation	-		---	---	ΔPo t	---	-25	%
2212	Pulse emission (2)	-		---	---	Δik	---	120	mA
---	Life test (2)	10	Group C; Ef = 6.0 V; F = 5 kHz ± 10%; Ebb = 1,000 Vdc; eg and Rk/Ik = 125 mAdc (min); Ic = 45 mAdc (min); TA = 250°C (min); t = 100 hours	---	---	---	---	---	---
---	Life-test (2) end point:								
2212	Pulse emission (2)	-		---	---	Δik	---	120	ma
---	Life-test (3)	9	Group S (weekly); Ef = 6.0 V; t = 500 hours; filament standby;	---	---	---	---	---	---
---	Life-test (3) end points:								
1236	Power oscillation	-		---	---	ΔPo t	---	-25	%
2212	Pulse emission (2)	-		---	---	Δik	---	120	ma
	Periodic-check tests								
1002	Barometric pressure, reduced	23	Pressure = 54 mmHg (max); voltage = 1,000 Vac; TA = 30° ± 10°C	---	---	---	---	---	---
(E)---	Variable-frequency vibration	13, 18, 24	F = 55 to 500 Hz; 10 G peak (min); Ec Ib = 10 mAdc; Rp = 10,000 ohms; Ebb = 300 Vdc	---	---	Ep	---	250	mVac
(E)1031	Vibration fatigue	13, 14, 26	Eb = Ec = 0; Ef = 6.0 V	---	---	---	---	---	---
---	Fatigue-test end point:								
1266	Total grid current	27		---	---	Ic	---	-10	μAdc
(E)---	Torque	13, 25, 28	No voltages	---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Periodic-check tests -Continued								
---	Torque-test end point:								
1266	Total grid current	-		---	---	Ic	---	-10	μAdc
(E) 1041	Shock (1)	12, 13, 14	400 G peak (min); duration = 0.5 ms (min); no voltages; use jig in accordance with Drawing 159-JAN	---	---	---	---	---	---
---	Shock-test (1) end point:								
1266	Total grid current	27		---	---	Ic	---	-10	μAdc
(E) 1041	Shock (2)	12, 13, 14, 18	60 G peak (min); duration = 11 ±2 ms; no voltages; use jig in accordance with Drawing 280-JAN	---	---	---	---	---	---
---	Shock-test (2) end point:								
1266	Total grid current	27		---	---	Ic	---	-10	μAdc

NOTES:

1. For grid-pulse operation, the maximum CW ratings shall apply.
2. The transit-time heating effect of the cathode shall be compensated for by a reduction in heater voltage after dynamic operation of the tube has started. The back heating is a function of frequency, grid current, grid bias, anode current, duty cycle, and circuit design and adjustment. There is an optimum heater voltage which will maintain the cathode at the correct operating temperature for a particular set of operating conditions. A maximum variation of ±5 percent from optimum is permitted. No reduction in heater voltage is required up to and including 500 MHz.
3. 1,200 volts is at the crest of the audio wave.
4. Up to 100 watts dissipation is allowable with forced-air cooling of the anode.
5. For 100 percent modulation including dissipation resulting from power supplied by the modulator.
6. The maximum instantaneous peak grid voltage for CW ratings shall be within the range of + 30 to -400 volts.
7. Maximum envelope temperature shall be limited to the specified maximum of 300°C under all operating conditions. Where emphasis is placed on long life and reliability, lower tube envelope temperatures shall be used.
8. Sufficient conduction or convection cooling shall be provided for all seals to limit temperatures in accordance with note 7. For anode cooling, an airflow of 12.5 cfm minimum shall be supplied at sea level, and with airflow at 25°C maximum directed with a cowl as shown on Drawing 157-JAN. Where long life and consistent performance are factors, cooling in excess of minimum requirements is normally beneficial. Unless otherwise specified, conduction cooling or an airflow of up to 12.5 cfm may be supplied for anode cooling in all electrical tests involving application of heater voltage.
9. Group S shall apply, except that it shall be permissible to represent a lot not to exceed one month's production with a sample consisting of four times the stated (weekly) sample, both for the multiple short life test and the sequential life test.

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NOTES: -Continued

10. Test in circuit in accordance with figure 3, or equivalent.

11. Life test shall be run in a self-excited oscillator cavity.

12. Each tube shall be subjected to a total of 15 shocks; that is, 5 shocks in each position X, Z+, and Z- in any sequence as shown on figure 1.

13. Other tube contact configurations may be used provided the tube contact area remains unchanged and the socket, jig, or cavity gives equal performance. Mounting of the jig, socket, or cavity may be at the option of the manufacturer.

14. Tubes subjected to destructive tests shall not be delivered on the contract or order.

15. All tests listed under quality conformance inspection, part 1, are to be performed at the conclusion of the holding period.

16. Test to be made in cavity in accordance with Drawing 160-JAN. The cavity shall be connected to a standard load with a VSWR less than 1.5. The output coupling from the oscillator and grid or cathode resistor, or both, may be adjusted for maximum power output.

17. Airflow through the anode radiator shall be between 3.5 and 4.0 cfm with cowl as shown on Drawing 157-JAN. The grid current shall be measured at not less than 5 minutes after anode dissipation has been set at 100 watts.

(F) 18. Perform test on 10 tubes selected at random from the first production lot of each calendar year. If more than one tube fails, the test shall become a part of quality conformance inspection, part 2, with AQL 6.5, inspection level S3, on all lots in process. After three consecutive successful submissions, the test shall revert to a 10-tube annual test.

19. The applied voltage pulse shape shall be measured with noninductive resistor of 1,150 ohms ±2 percent inserted in place of the tube. The pulse shape shall be tp = 3.0 μs ±10 percent, tr = 0.4 μs maximum, tf = 0.7 μs maximum. The pulse repetition rate shall be adjusted so that Du = 0.0025 ±5 percent with the above measured pulse length. Test in cavity in accordance with Drawing 279-JAN. The cavity shall be connected to a load with a VSWR less than 1.5. The oscillator output coupling and grid or cathode resistor may be adjusted for maximum power output.

20. Measured in socket in accordance with Drawing 158-JAN.

21. Grid-anode resonance: Test in cavity in accordance with Drawing 278-JAN. Cavity shall resonate at 1.354 GHz ±2.0 MHz with tuning slug in accordance with Drawing 277-JAN at TA = 25° ±5°C.

    Grid-cathode resonance: Test in cavity in accordance with Drawing 283-JAN. Cavity shall resonate at 1.719 GHz ±2.0 MHz with tuning slug in accordance with Drawing 277-JAN at TA = 25° ±5°C.

    When plotted on graphs of resonant frequency versus grid-anode capacitance and resonant frequency versus grid-cathode capacitance, the tube under test shall be represented by a point within a parallelogram whose four corners are located by the following points:

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Points	Capacitance (pF)		Frequency (GHz)	
	C-gp	C-gk	F-gp	F-gk
1	1.95	5.6	1.955	1.784
2	1.95	5.6	1.985	1.814
3	2.15	7.0	1.900	1.751
4	2.15	7.0	1.930	1.781

22. Test shall be made in cavity in accordance with Drawing 281-JAN. Calibrate the circuit by adjusting the cavity until the frequency of resonance is 2.0 GHz ±5 MHz at room temperature, with a tube which meets the resonance test requirements. In the grid-distortion test, the frequency shall first be measured with Ef = 6.0 volts, then the grid dissipation increased to 2 watts with Eb = -150 Vdc. The change in frequency shall be not greater than the limit specified herein. Cooling air at room temperature may be used.

23. Voltage shall be 60 Hz ac applied between anode and grid. No other voltages shall be applied. There shall be no evidence of failure as indicated by an arc-over. Perform test on 10 tubes selected from the first production lot of each year. If more than one tube fails, the test shall become a part of quality conformance inspection, part 2, with AQL 6.5, inspection level S3. After three consecutive successful submissions the test shall revert to a 10-tube annual test. This is not a destructive test.

24. The tube shall be mounted in socket in accordance with Drawing 276-JAN and vibrated with simple harmonic motion. The peak acceleration over the frequency range shall be within ±20 percent of the reference acceleration at 100 Hz. The frequency shall vary from 55 to 500 Hz and return to 55 Hz with approximately logarithmic progression and shall require 4 minutes minimum, 6 minutes maximum to traverse the range. Each tube shall be vibrated for 30 minutes in each position X and Z, except that if the cumulative result of tests on 50 or more tubes of a construction show that more than 75 percent of the tubes have higher output voltages in one position, subsequent measurements need to be taken only in the position giving the higher reading. The voltages specified herein shall be applied to the tube during vibration. The value of the alternating voltage, Ep, produced across the resistor, Rp, as a result of vibration shall be measured with a suitable device. This device shall have an appropriate voltage range, shall have the ability to measure, with an error of less than 10 percent, the rms value of a sine wave of voltage at all frequencies from 20 to 20,000 Hz. The value of the vibrational output, Ep, shall not exceed the limit specified herein at any point in the sweep-frequency range during the last complete cycle of cycling vibration.

(E) 25. Test 10 tubes selected at random from the first production lot of each calendar year and approximately every 90 days during the year. If more than one tube fails to pass the specified end points, the failed test shall become a part of quality conformance inspection, part 2, with AQL 6.5, inspection level S3, on all lots in process. After three consecutive successful submissions, the test shall revert to a 10-tube quarterly test. This is a nondestructive test, except in case of failure.

26. Test four tubes selected at random from the first production lot of each year. If more than one tube fails to pass the specified end points, the failed test shall become a part of quality conformance inspection, part 2, with AQL 6.5, inspection level S3, on all lots in process. After three consecutive submissions, the test shall revert to a 4-tube annual test. This is a destructive test.

27. The following does not apply: Tubes which show permanent or tap shorts or open circuits following the applicable test.

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(E) 28. Torque test shall be performed as follows:

(a) The tube shall be held securely at the cathode connection. A force of 5 pounds shall be applied to the heater cup without perceptible shock. This test may be made by applying the force at right angles to the inside of the cup at a point 7/64 (2.78 mm) ±1/64 (0.40 mm) inch from the cathode end of the tube. An approved equivalent method may be used. The heater cup shall not loosen or short circuit on the cathode connection.

(b) A torque of 15 inch-pounds shall be applied between anode and cathode without shock.

(c) A torque of 40 inch-pounds shall be applied between anode and grid without shock.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 11, 17
DSA - ES

User activities:
Army - ME, WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Army - EL

Agent: DSA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	1.815 (46.10)	1.875 (47.63)
B		1.534 (38.96)
C		1.475 (37.47)
D	1.289 (32.74)	1.329 (33.76)
F	.880 (22.35)	.920 (23.37)
G	.462 (11.73)	.477 (12.12)
J	.766 (19.46)	.826 (20.98)
N	1.025 (26.04)	1.035 (26.29)
R	.655 (16.64)	.665 (16.89)
T	.213 (5.41)	.223 (5.66)
U	.315 (8.00)	.325 (8.26)
Quality conformance inspection, part 3 (periodic check) (see note b)		
E	1.085 (27.56)	1.135 (28.83)
H		.040 (1.02)
I	.125 (3.18)	.185 (4.70)
K	.025 (.64)	.046 (1.17)
L	1.234 (31.34)	1.264 (32.11)
M	1.180 (29.97)	1.195 (30.35)
P	.772 (19.61)	.792 (20.12)
Q	.541 (13.74)	.561 (14.25)
S		.545 (13.84)
V		.086 (2.18)
W		.100 (2.54)
X		.035 (.89)
Y	.105 (2.67)	.145 (3.68)
Z	.650 (16.51)	.850 (21.59)
Electrode contact areas (see note n)		
AA	.035 (.89)	.361 (9.17)
AB	1.185 (30.10)	1.265 (32.13)
AC	1.534 (38.96)	1.728 (43.89)
AD	1.475 (37.47)	1.815 (46.10)

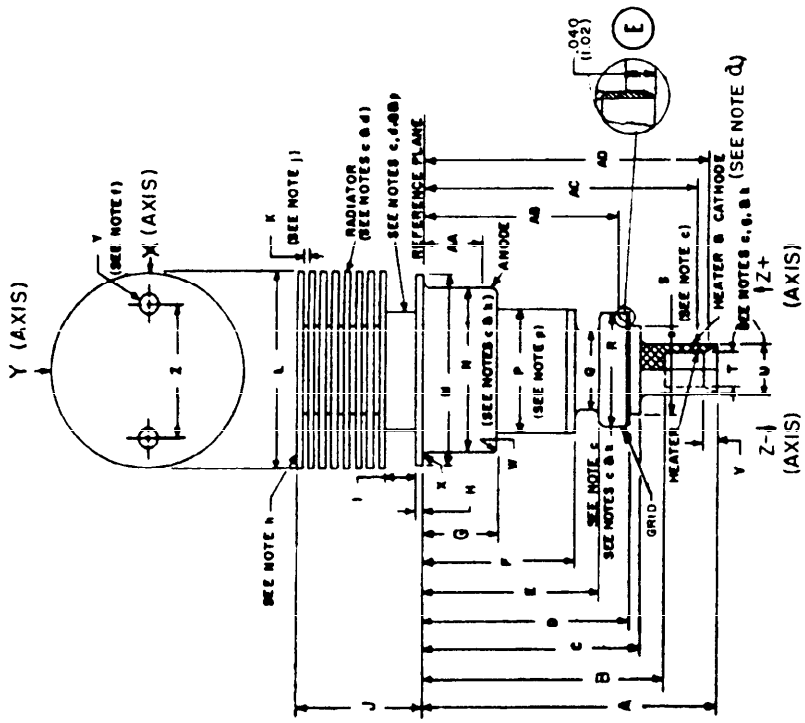


FIGURE 1. Outline drawing of electron tube type 7289.

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NOTES:

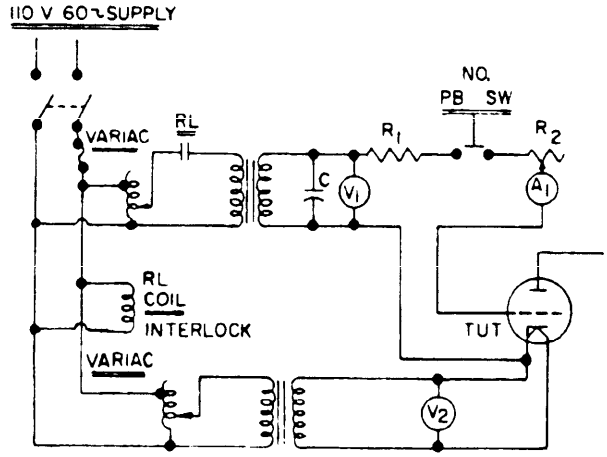
Ⓔ a. Insulation material is required between heater and heater-cathode and shall be securely affixed.
b. These dimensions shall be tested on 10 tubes per month when in continuous production. Failure of more than one tube to meet tolerances for any dimension shall cause that dimension to become, for all lots in process, part of quality conformance inspection, part 2.
c. Silver plated 30 MSI minimum. Note b shall apply.
d. Plating not required over radiator and radiator support of copper, aluminum, or approved equivalent.
e. This surface shall be used for measurement of anode shank temperature.
f. Holes for tube extractor through top fin only.
g. Inner edge of heater and outer edge of cathode rf connection shall be free from burrs and sharp edges.
h. This fin shall withstand a 6-inch drop test without loosening and without distortion as judged by ability to maintain dimension K. Note b shall apply.
j. Distortion of fins permissible provided distance between adjacent fins at any point on circumference meets tolerances for dimension K.
k. Eccentricity of contact surfaces shall be gaged from center line of reference and shall be as follows. Note b shall apply.

Contact surface	TIR maximum	Reference
Anode	.020(.51 mm)	Cathode
Grid	.020(.51 mm)	Cathode
Heater	.012(.30 mm)	Cathode

m. Diameters N, R, T, and U shall apply throughout entire contact areas as defined by dimensions AA, AB, AC, and AD, respectively.
n. Dimensions in electrode contact areas table are for socket design purposes and are not intended for inspection purposes.
p. This surface shall not be used for clamping or locating.

FIGURE 1. Outline drawing of electron tube type 7289 - Continued.

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C – 1 μF, 1,000 Vdc
R1 – 500 ohms, 10 W
R2 – 1,000 ohms, 50 W, variable
V1 – 0 to 400 Vac
V2 – 0 to 10 Vac
A1 – 0 to 150 mAdc

Test procedure:

a. Remove tube from preheat rack and place in test socket.
b. Check heater voltage at V2 (6.0 Vac) and grid voltage at V1 (285 Vac).
c. Press pushbutton switch, applying voltage between grid and cathode. Pushbutton switch is normally open.
d. Adjust R2 to 100 mAdc on A1 if necessary.
e. Hold pushbutton switch closed for at least 10 seconds. Tube shall be rejected if current drops to zero or fluctuates and continuous arcing occurs.

FIGURE 2. Grid arc test set wiring diagram.

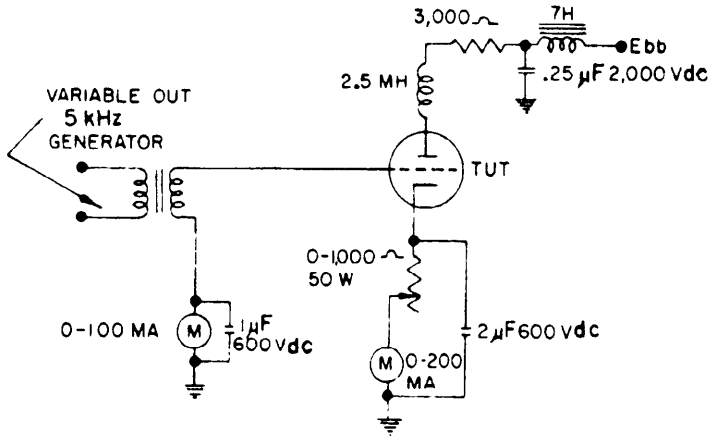


FIGURE 3. 5 kHz life-test circuit.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-E-1/1120E ELECTRON TUBE,NEGATIVE GRID(MICROWAVE) TYPE 7289

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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